

BOSE CONDENSATION IN BIOSYSTEMS

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Received 14 September 1977

A microscopic approach to Fröhlich's theory of Bose condensation in biological systems is discussed. The conclusions from this microscopic analysis offer further support for Fröhlich's hypothesis.

The possible importance of vibrational modes in the frequency range of 10^{11} – 10^{12} sec $^{-1}$ of biological macromolecules in stimulating important life processes was first suggested by Fröhlich [1–5]. Recently, several experimental results have been published which tend to support Fröhlich's theory [6–10].

Generally, biological systems possess constituents that are capable of electric dipolar oscillations. Long range Coulomb interaction between these dipoles establishes a branch or branches of longitudinal electric modes. Fröhlich has suggested if energy is supplied locally to one constituent (or removed from it), this energy variation will be shared by all the constituents within the branch of vibrational modes due to the long range coulomb interaction. He further concluded if energy is supplied beyond a certain critical rate, the energy redistribution among the dipolar modes will lead to its concentration in the lowest energy state (i.e. Bose–Einstein condensation).

Fröhlich's mathematical analysis was based upon the utilization of rate equations. We have approached the theoretical problem using microscopic techniques and the interaction Hamiltonian. As Fröhlich, we suppose that the biological system consists of the following:

- (i) oscillating segments of giant dipoles occurring along macromolecules,
- (ii) the remainder of the biosystem which forms a heat bath,
- (iii) an external energy source which couples to the oscillating units.

The interaction between oscillating units produces a narrow band of frequencies, ω_i ($\omega_0 \leq \omega_i \leq \omega_{\max}$) representing the normal electromagnetic modes. Assigned to each mode are creation and destruction operators, a_i^+ and a_i . The vibrational modes of the heat bath, Ω_k , are associated with creation and destruction operators, b_k^+ and b_k . Similarly, the external energy source has frequencies, $\bar{\Omega}_p$, and creation and destruction operators P_p^+ and P_p associated with it. Thus, the Hamiltonian of the biosystem can be given as:

$$H = \sum_i \hbar \omega_i a_i^+ a_i + \sum_k \hbar \Omega_k b_k^+ b_k + \sum_p \hbar \bar{\Omega}_p P_p^+ P_p + \hbar \sum_{i,k} (\lambda b_k^+ a_i + \lambda^* b_k a_i^+) + \frac{1}{2} \hbar \sum_{i,j,k} (\chi a_i^+ a_i b_k^+ + \chi^* a_j a_i^+ b_k) + \hbar \sum_{p,i} (\xi P_p a_i^+ + \xi^* P_p^+ a_i), \quad (1)$$

where λ , χ and ξ are coupling constants for the one quantum process, the two quantum process and the energy supply to the oscillating constituents respectively. Using the commutation relations the rate of change of i th mode quanta is

$$\dot{n}_i = (i\hbar)^{-1} [n_i, H], \quad (2)$$

where $n_i = a_i^+ a_i$.

Using the commutation relations for the various creation and destruction operators, we found the expectation value of $\dot{n}_i$ ($\langle \dot{n}_i \rangle = \langle \psi(t) | \dot{n}_i | \psi(t) \rangle$) to be:

$$\langle \dot{n}_i \rangle = S - \phi(T, \omega_i) [\langle n_i \rangle \exp(\hbar\omega_i/kT) - (1 + \langle n_i \rangle)] - \sum_j \Lambda(T, \omega_i, \omega_j) [\langle n_i \rangle (1 + \langle n_j \rangle) \exp(\hbar(\omega_i - \omega_j)/kT) - \langle n_j \rangle (1 + \langle n_i \rangle)], \quad (3)$$

where the angular brackets refer to thermal averaging in the grand ensemble appropriate to the entire system.

$|\psi(t)\rangle$ is the state vector of the whole system at time t . Since $\langle \dot{n}_i \rangle$ cannot be solved exactly, perturbation theory was utilized to obtain it to lowest order in coupling strength. For stationarity, $\langle \dot{n}_i \rangle = 0$. As a result, we found the average number of i th mode quanta to be:

$$\langle n_i \rangle = \left[1 + \frac{S}{\phi(T, \omega_i) + \sum_j \Lambda(T, \omega_i, \omega_j) \langle n_j \rangle} \right] \{ \exp[\hbar\omega_i - \mu/kT] - 1 \}^{-1}, \quad (4)$$

where

$$\exp(-\mu/kT) = \frac{1 - \sum_j \Lambda(T, \omega_i, \omega_j) (\langle n_j \rangle - (1 + \langle n_j \rangle) \exp(-\hbar\omega_j/kT))}{\phi + \sum_j \Lambda \langle n_j \rangle}, \quad (5)$$

$$S = 4\pi |\xi|^2 \sum_i \langle p_i^+ p_i \rangle, \quad (6)$$

$$\phi(T, \omega_i) = 4\pi |\lambda|^2 \exp(-\hbar\omega_i/kT) [1 + N(\hbar\omega_i)], \quad (7)$$

and

$$\Lambda(T, \omega_i, \omega_j) = \pi |\chi|^2 \exp[-\hbar(\omega_i - \omega_j)/kT] \times [N(\hbar\omega_j - \hbar\omega_i)], \quad \omega_j > \omega_i, \\ \times [1 + N(\hbar\omega_i - \hbar\omega_j)], \quad \omega_i > \omega_j, \quad (8)$$

where $N(\hbar\omega)$ is the number of excitations with energy $\hbar\omega$ in the heat bath.

Eqs. (4) and (5) are of exactly the same form as those derived by Fröhlich. Bose–Einstein condensation into the lowest energy state occurs when μ approaches $\hbar\omega_0$. This takes place when the energy supplied exceeds a certain critical rate as shown by Fröhlich. Fröhlich formed the ansätze in his rate equations by requiring a Bose–Einstein distribution for thermal equilibrium when no external power is supplied. Otherwise, the form of the rate equations were rather arbitrary. However, by approaching Fröhlich's theory from a macroscopic point of view, we have arrived at the same conclusions as Fröhlich did and removed the arbitrariness. Our results, therefore, further strengthen the possibility that such a phenomenon as Bose condensation in biological systems exists.

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